

LABORATORY RECORD

Course: Full Stack Web Development

Course Code: 23CM4121

Branch: CSM

Section: B

Task No.: 3

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1. TITLE

Design a Responsive Webpage Using HTML5 and CSS3

2. AIM

To design and develop a modern, responsive webpage using **HTML5 and CSS3** with a navigation bar, hero section, about section, services section, contact form, and footer.

3. OBJECTIVE

The objectives of this experiment are:

- To understand the basic structure of an HTML5 webpage.
 - To design a webpage using HTML5 semantic elements.
 - To apply CSS3 styling to HTML elements.
 - To create a responsive navigation bar.
 - To design an attractive hero section using CSS.
 - To create service cards using CSS Flexbox.
 - To design and style a contact form.
 - To implement hover effects and smooth scrolling.
 - To make the webpage responsive using CSS media queries.
 - To understand the separation of HTML structure and CSS presentation.
-

4. THEORY

HTML5 provides the structural elements of a webpage — headings, sections, forms, and text content.

CSS3 is used to style and lay out that structure — colours, spacing, fonts, and responsive behaviour — while keeping the HTML file focused purely on content and the CSS file focused purely on presentation. Keeping them separate makes a site easier to maintain and reuse.

Several CSS3 features make this page both attractive and responsive without any JavaScript.

- **Flexbox** (**display: flex**) arranges the navigation links and the service cards in a row that wraps neatly onto new lines when space runs out.
- **Media queries** (**@media**) apply different CSS rules at different screen widths, which is the core mechanism behind responsive design.
- **Transitions and :hover** create smooth colour and shadow changes when the user points at a link, button, or card.
- **scroll-behavior: smooth** makes in-page anchor links (like the nav bar) scroll smoothly instead of jumping instantly.
- **Box model properties** (padding, margin, border-radius, box-shadow) control spacing and give the cards and buttons their rounded, elevated look.

The basic flow of the page is:

Browser loads index.html → Browser links style.css → CSS rules are applied to HTML elements → Layout adjusts to screen width via media queries

5. ALGORITHM / PROCEDURE

1. Create two files: index.html and style.css, and link the CSS file inside the HTML head.
2. Reset default margins/padding and set box-sizing to border-box for consistent sizing.
3. Build a header with a logo and a navigation list linking to in-page sections.
4. Create a hero section with a heading, a description, and a call-to-action button.
5. Create an about section with a heading and a descriptive paragraph.
6. Create a services section and arrange three service cards using CSS Flexbox.
7. Style each card with padding, rounded corners, and a shadow that lifts on hover.
8. Create a contact form with labelled text, email, and message fields.
9. Style the form inputs with borders, rounded corners, and a focus colour change.
10. Create a footer with a centred copyright notice.
11. Add hover effects to links, buttons, and cards using the :hover pseudo-class.
12. Add scroll-behavior: smooth so navigation links scroll smoothly to their section.
13. Add media queries at 768px and 480px breakpoints to adjust layout for tablets and phones.
14. Open index.html in a browser and resize the window to verify responsive behaviour.

6. PROGRAM

index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>BrightWave Solutions</title>
<link rel="stylesheet" href="style.css">
</head>
```

```
<body>

<!-- ===== HEADER / NAV ===== -->
<header>
  <nav>
    <div class="logo">BrightWave</div>
    <ul class="nav-links">
      <li><a href="#home">Home</a></li>
      <li><a href="#about">About</a></li>
      <li><a href="#services">Services</a></li>
      <li><a href="#contact">Contact</a></li>
    </ul>
  </nav>
</header>

<!-- ===== HERO SECTION ===== -->
<section class="hero" id="home">
  <h1>Build Something Amazing</h1>
  <p>We craft modern, responsive web experiences that help your business grow and connect with the right audience.</p>
  <a href="#contact" class="btn">Get Started</a>
</section>

<!-- ===== ABOUT SECTION ===== -->
<section class="about" id="about">
  <h2>About Us</h2>
  <p>BrightWave Solutions is a digital studio specializing in responsive web design. Our team combines creativity and technology to build websites that look great and perform even better on every device – from desktops to smartphones.</p>
</section>

<!-- ===== SERVICES SECTION ===== -->
<section class="services" id="services">
  <h2>Our Services</h2>
  <div class="service-cards">
    <div class="card">
      <h3>Web Design</h3>
      <p>Clean, modern designs tailored to your brand identity and audience.</p>
    </div>
    <div class="card">
      <h3>Responsive Development</h3>
      <p>Websites that adapt seamlessly across mobiles, tablets, and desktops.</p>
    </div>
    <div class="card">
      <h3>Performance Optimization</h3>
      <p>Fast-loading pages built with clean, efficient HTML5 and CSS3.</p>
    </div>
  </div>
</section>

<!-- ===== CONTACT FORM ===== -->
<section class="contact" id="contact">
  <h2>Contact Us</h2>
  <form>
    <label for="name">Name</label>
    <input type="text" id="name" name="name" placeholder="Your name" required>
  </form>
</section>
```

```

        <label for="email">Email</label>
        <input type="email" id="email" name="email" placeholder="you@example.com" required>

        <label for="message">Message</label>
        <textarea id="message" name="message" rows="5" placeholder="Write your message..."
required></textarea>

        <button type="submit" class="btn">Send Message</button>
    </form>
</section>

<!-- ===== FOOTER ===== -->
<footer>
    <p>&copy; 2026 BrightWave Solutions. All rights reserved.</p>
</footer>

</body>
</html>

```

style.css

```

/* ===== RESET & BASE ===== */
* {
    margin: 0;
    padding: 0;
    box-sizing: border-box;
}
html {
    scroll-behavior: smooth; /* smooth scrolling for nav links */
}
body {
    font-family: Arial, sans-serif;
    line-height: 1.6;
    color: #333;
}
a {
    text-decoration: none;
    color: inherit;
}
ul {
    list-style: none;
}

/* ===== NAVIGATION BAR ===== */
header {
    background-color: #1f2937;
}
nav {
    display: flex;
    justify-content: space-between;
    align-items: center;
    max-width: 1100px;
    margin: 0 auto;
    padding: 15px 20px;
}

```

```

    flex-wrap: wrap;
}
.logo {
    color: #fff;
    font-size: 1.5rem;
    font-weight: bold;
}
.nav-links {
    display: flex;
    gap: 25px;
}
.nav-links a {
    color: #e5e7eb;
    font-weight: 500;
}
.nav-links a:hover {
    color: #38bdf8; /* hover effect */
}

/* ===== HERO SECTION ===== */
.hero {
    background-color: #1f2937;
    color: #fff;
    text-align: center;
    padding: 100px 20px;
}
.hero h1 {
    font-size: 2.5rem;
    margin-bottom: 15px;
}
.hero p {
    font-size: 1.1rem;
    max-width: 600px;
    margin: 0 auto 30px;
    color: #d1d5db;
}
.btn {
    display: inline-block;
    background-color: #38bdf8;
    color: #1f2937;
    padding: 12px 30px;
    border-radius: 30px;
    font-weight: bold;
    border: none;
    cursor: pointer;
    font-size: 1rem;
}
.btn:hover {
    background-color: #0ea5e9;
}

/* ===== ABOUT SECTION ===== */
.about {
    max-width: 900px;
    margin: 0 auto;
    padding: 80px 20px;
}

```

```

    text-align: center;
}
.about h2 {
    font-size: 2rem;
    margin-bottom: 20px;
    color: #1f2937;
}
.about p {
    color: #4b5563;
}

/* ===== SERVICES SECTION (Flexbox cards) ===== */
.services {
    background-color: #f3f4f6;
    padding: 80px 20px;
    text-align: center;
}
.services h2 {
    font-size: 2rem;
    margin-bottom: 50px;
    color: #1f2937;
}
.service-cards {
    display: flex;
    flex-wrap: wrap;
    justify-content: center;
    gap: 30px;
    max-width: 1100px;
    margin: 0 auto;
}
.card {
    background-color: #fff;
    border-radius: 12px;
    padding: 35px 25px;
    flex: 1 1 280px;
    max-width: 320px;
    box-shadow: 0 4px 10px rgba(0, 0, 0, 0.08);
}
.card:hover {
    box-shadow: 0 8px 16px rgba(0, 0, 0, 0.12); /* hover effect */
}
.card h3 {
    margin-bottom: 12px;
    color: #1f2937;
}
.card p {
    color: #6b7280;
    font-size: 0.95rem;
}

/* ===== CONTACT FORM ===== */
.contact {
    max-width: 600px;
    margin: 0 auto;
    padding: 80px 20px;
}

```

```

.contact h2 {
  text-align: center;
  font-size: 2rem;
  margin-bottom: 30px;
  color: #1f2937;
}

.contact form {
  display: flex;
  flex-direction: column;
  gap: 15px;
}

.contact label {
  font-weight: 600;
  color: #374151;
}

.contact input,
.contact textarea {
  width: 100%;
  padding: 12px;
  border: 1px solid #d1d5db;
  border-radius: 8px;
  font-family: inherit;
}

.contact input:focus,
.contact textarea:focus {
  outline: none;
  border-color: #38bdf8;
}

.contact button {
  align-self: center;
}

/* ===== FOOTER ===== */
footer {
  background-color: #1f2937;
  color: #d1d5db;
  text-align: center;
  padding: 25px 20px;
}

/* ===== RESPONSIVE MEDIA QUERIES ===== */
@media (max-width: 768px) {
  nav {
    flex-direction: column;
    gap: 10px;
  }

  .nav-links {
    flex-wrap: wrap;
    justify-content: center;
    gap: 15px;
  }

  .hero h1 {
    font-size: 2rem;
  }
}

```

```

@media (max-width: 480px) {
  .hero {
    padding: 60px 15px;
  }
  .about, .services, .contact {
    padding: 50px 15px;
  }
  .card {
    flex: 1 1 100%;
  }
}

```

7. CODE EXPLANATION

Code	Explanation
<header>, <nav>	Hold the site logo and the navigation links at the top of the page.
<section id="...">	Marks the hero, about, services, and contact blocks so nav links can jump to them.
<div class="service-cards">	A Flexbox container that lays the three service cards out in a row.
<form>, <label>, <input>, <textarea>	Build the contact form and its labelled fields.
required	HTML5 built-in validation that blocks submission if a field is left empty.
* { box-sizing: border-box; }	Makes padding and borders count inside an element's declared width, avoiding layout surprises.
display: flex; flex-wrap: wrap;	Arranges nav links and service cards in a row that wraps onto new lines when needed.
:hover	Applies a style only while the mouse pointer is over an element, used for links, buttons, and cards.
scroll-behavior: smooth;	Makes clicking a nav link scroll smoothly to its section instead of jumping instantly.
box-shadow, border-radius	Give the cards, buttons, and inputs their soft, rounded, elevated appearance.
@media (max-width: 768px)	Applies alternate CSS rules on tablets and smaller screens.

@media (max-width: 480px)	Applies further adjustments for small mobile screens.
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8. TEST CASES AND EXECUTION DATA

The page was opened in a browser at different window widths, and each interactive element was tested; the results were recorded below.

Test Case	Action	Expected Result	Actual Result	Status
TC-01	Open index.html in a browser	Page loads with header, hero, about, services, contact, and footer	Page displayed correctly	Pass
TC-02	Click 'Services' in the navigation bar	Browser scrolls smoothly to the Services section	Smooth scroll worked correctly	Pass
TC-03	Hover over a service card	Card lifts slightly with a stronger shadow	Hover effect displayed	Pass
TC-04	Leave the Name field empty and click Send Message	Browser blocks submission and prompts for the field	Required-field prompt shown	Pass
TC-05	Enter an invalid email format	Browser shows an email-format validation error	Validation error displayed	Pass
TC-06	Resize browser window to tablet width ($\leq 768\text{px}$)	Nav links wrap and center; layout stays readable	Layout adjusted correctly	Pass
TC-07	Resize browser window to mobile width ($\leq 480\text{px}$)	Service cards stack in a single column	Cards stacked correctly	Pass

Additional Validation Test

Test Case	Input/Action	Expected Result	Status
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TC-08	Click into the Name input field	Border colour changes to blue on focus	Pass
TC-09	Click 'Get Started' hero button	Browser scrolls smoothly to the Contact section	Pass

9. OUTPUT

Output 1: Header, Navigation, and Hero Section (Desktop)



Output 2: About and Services Sections (Desktop)



About Us

BrightWave Solutions is a digital studio specializing in responsive web design. Our team combines creativity and technology to build websites that look great and perform even better on every device — from desktops to smartphones.

Our Services

Web Design

Clean, modern designs tailored to your brand identity and audience.

Responsive Development

Websites that adapt seamlessly across mobiles, tablets, and desktops.

Performance Optimization

Fast-loading pages built with clean, efficient HTML5 and CSS3.

Output 3: Contact Form and Footer (Desktop)

Contact Us

Name

Email

Message

Send Message

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Output 4: Full Page on a Mobile Screen – Responsive Layout

BrightWave

[Home](#) [About](#) [Services](#) [Contact](#)

Build Something Amazing

We craft modern, responsive web experiences that help your business grow and connect with the right audience.

[Get Started](#)

About Us

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Our Services

Web Design

Clean, modern designs tailored to your brand identity and audience.

Responsive Development

Websites that adapt seamlessly across mobiles, tablets, and desktops.

10. RESULT

Thus, a modern, responsive webpage was successfully designed and developed using HTML5 and CSS3. The navigation bar, hero section, about section, service cards, contact form, and footer were implemented with hover effects and smooth scrolling, and the layout was verified to adapt correctly across desktop, tablet, and mobile screen widths using CSS media queries.